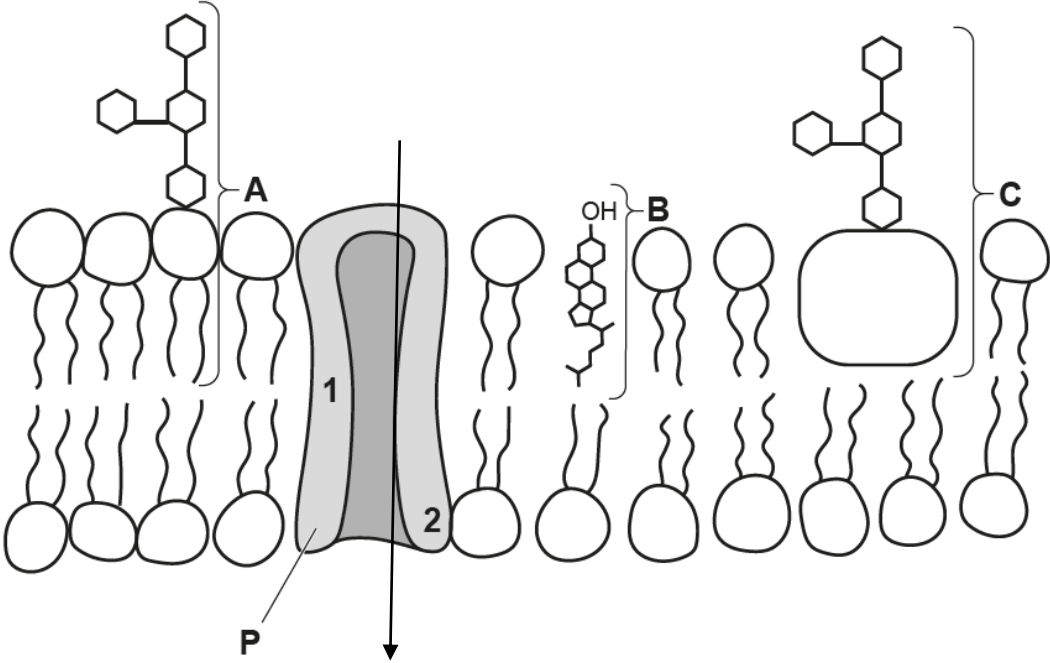


1(a)	A glycolipid ; B cholesterol ; C glycoprotein ;	3
1(b)	 The diagram illustrates a cross-section of a cell membrane. It features a phospholipid bilayer with hydrophilic heads (circles) and hydrophobic tails (wavy lines). A large integral protein is shown spanning the membrane. A vertical arrow passes through the protein, with the number '1' in the upper part and '2' in the lower part. A label 'P' with a line points to the phospholipid heads. On the left, a glycolipid (A) is shown with a branched carbohydrate chain. In the center, a cholesterol molecule (B) is depicted with its characteristic four-ring structure and a hydroxyl group (OH). On the right, a glycoprotein (C) is shown with a large protein head and a branched carbohydrate chain. Brackets group these three structures as A, B, and C respectively. 1 arrow through protein ; 2 direction of arrow correct ;	2

Question	Answer	Marks
1(c)	<i>any three from:</i> fluidity (of membrane) increases ; unsaturated fatty acids have double bonds ; (double bonds) cause the fatty acid chain to, kink / bend / AW ; unsaturated fatty acids cannot lie as close together as saturated fatty acids A ora or <i>ref. to</i> increased distance between phospholipids / AW ; <i>ref. to</i> weakening the hydrophobic interactions between, fatty acid chains / cholesterol ;	3
1(d)	<i>any one from:</i> position 1 has amino acids with, hydrophobic / non polar / AW, R-groups ; position 2 has amino acids with, hydrophilic / polar / ionic / AW, R-groups; A ref. to amine and carboxyl groups as appropriate	1

Question	Answer	Marks
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